Conjecture 00000001042: an exact affine variance and the fixed-degree asymptotic obstruction

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1 Statement, interpretation, and scope

The source conjecture says: "The variance of the value-set size of a random polynomial f uniform of degree $\leq d$ is asymptotically $q(1-1/e)/d$." The Chinese version likewise specifies degree at most d. The source is conjecture 00000001042 at upstream commit: `b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

We disprove the reading in which q tends to infinity through finite-field cardinalities while d is fixed, with the stated class including $d = 1$. The source does not specify a limiting regime. We do not claim to settle a different formulation restricted to growing d, or one excluding $d = 1$. Uniform means uniform on all coefficient vectors, including zero leading coefficients and the zero polynomial. In degree at most one this is the uniform distribution on the q^2 distinct formal polynomials $aX + b$.

2 Exact computation for every finite field

Let K be any finite field of cardinality q, and let (a, b) be uniform on K^2. Define the actual value-set size

$$V(a, b) = \#\{ax + b : x \in K\}.$$

If $a = 0$ the image is the singleton $\{b\}$, so $V = 1$. If $a \neq 0$, the inverse image of y is $(y - b)/a$; hence the map is surjective and $V = q$. There are q pairs with zero slope and $q(q-1)$ with nonzero slope. Thus

$$\mathbb{E}[V^k] = \frac{1 + (q-1)q^k}{q}, \qquad \mathbb{E}[V] = q - 1 + \frac{1}{q}.$$

The variance, equivalently the uniform mean squared deviation, is

$$\mathrm{Var}(V) = \mathbb{E}[V^2] - \mathbb{E}[V]^2 = \frac{(q-1)^3}{q^2}. \tag{1}$$

This is an identity for every finite field, not a finite numerical test.

3 Asymptotic contradiction

Let p_n enumerate the primes in increasing order, and take $K_n = \mathbb{Z}/p_n\mathbb{Z}$. These are finite fields and $p_n \to \infty$. Equation (1) gives

$$\frac{\mathrm{Var}_{K_n}(V)}{p_n} = \left(1 - \frac{1}{p_n}\right)^3 \longrightarrow 1.$$

But $\exp(1) > 1$, so $0 < 1 - 1/\exp(1) < 1$. The proposed fixed-$d = 1$ equivalence would instead imply a normalized limit $1 - 1/\exp(1)$. Uniqueness of limits yields a contradiction. In Lean we additionally refute the exact ratio formulation

$$\frac{\mathrm{Var}_{K_n}(V)}{p_n(1 - 1/\exp(1))} \longrightarrow 1.$$

Failure along this legitimate unbounded sequence already disproves the asserted q-asymptotic reading.

4 Formalization and semantic correspondence

The standalone project uses Lean 4.33.0 and the pinned Mathlib revision in the project configuration and dependency manifest. The main theorems in `lean/Main.lean` are:

- `affineEquiv`: an explicit bijection between coefficient pairs and all formal polynomials of natural degree at most one; `affine_eval` connects their evaluations to $ax + b$. Thus the uniform coefficient enumeration is exactly the requested uniform polynomial ensemble.
- `valueCount_zero`, `valueCount_ne_zero`, and `sum_power`: actual finite image cardinalities and exact sums.
- `variance_is_mean_squared_deviation` and `variance_formula`: the standard variance definition and (1), with no assumed moment or counting formula.
- `normalized_variance_limit`: the true limit along actual prime fields. The final theorem `refutes_asymptotic_equivalence` proves failure of the conjectured ratio limit.

The only foundational axioms allowed are Lean's usual `propext`, `Classical.choice`, and `Quot.sound`. There is no admitted proof, native decision shortcut, or new axiom. Reproduction commands and exact verification receipts accompany this report.

5 Previous submission and what is repaired

PR 155 already presented this elementary mathematical obstruction. It was closed unmerged: its Lean proof checked only $q = 101$, while the general identity and the limiting argument remained in prose. A finite discrepancy alone does not disprove an asymptotic claim. The maintainer's review identified both missing formal steps and the unspecified limiting regime. This submission credits that prior observation, proves the general finite-field identity and the limiting contradiction in Lean, and explicitly limits its conclusion to the fixed-degree, growing-field interpretation. It makes no claim to new mathematics or to resolving every possible alternative interpretation of the incomplete source statement.

6 Provenance

OpenAI Codex performed the reasoning, Lean formalization, and report preparation for the submitting account. A separate supervising Codex agent read the original statement and full proof source for semantic review; this is an AI review, not a human certification. The official competition review and acceptance remain the maintainers' responsibility.